COMP9602 — Convex Optimization

Assignment 1 (12%) [Learning Outcome 1]

Due by: 23:59, Sunday, October 4, 2026

Submission: please submit a softcopy through Moodle following the steps below:

1. Login Moodle.
2. Find "Assignments" in the left column and click "Assignment 1".
3. Click "Add submission", browse your solution file and save it. Done.
4. You will receive an automatic confirmation email if the submission was successful.
5. You can "Edit submission" to your already submitted file, but ONLY before the deadline.
6. Only .pdf files are accepted, and at most one file is allowed to be uploaded.

Total mark is 100.

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1. (10 marks)

Prove whether each of the following sets is convex or not:

(a) $\{(x, y) \in \mathbb{R}^2 : xy \leq 0\} \cap \{(x, y) \in \mathbb{R}^2 : y \geq 3x\}$

(b) $\{(x, y) \in \mathbb{R}^2_+ : e^{2y} \geq e^{\sqrt{2x}}\}$

2. (15 marks)

Let $n \geq 2$. Fix a vector $r \succ 0, r \in \mathbb{R}^n$ (i.e. $r_i > 0$ for all i) and define

$$C = \big\{A \in \mathbf{S}^n : x^T A x \geq 0 \ \text{ for all } x \in \mathbb{R}^n \text{ with } r_i \leq x_i \leq 3r_i,\ i = 1, \ldots, n\big\}.$$

Prove whether each statement below is true or false.

(a) C is a convex cone.

(b) $\mathbf{S}^n_+ \subseteq C$ (every positive semidefinite (PSD) matrix belongs to C).

3. (15 marks)

Let

$$f(x, t) = -\log\big(t^2 - 2x^T x\big), \qquad \operatorname{dom} f = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} : t > \sqrt{2}\,\|x\|_2\}.$$

(a) Show that $h(x, t) = t - \dfrac{2x^T x}{t}$ is concave on $\operatorname{dom} f$.

(b) Show that $-\log\big(t - \frac{2x^T x}{t}\big)$ is convex on $\operatorname{dom} f$.

(c) Show that f is convex on $\operatorname{dom} f$.

4. (20 marks)

Show that each of the following functions is convex.

(a) For $x \in \mathbb{R}^n$ with $n \geq 2$, define $q_x(t) = \sum_{i=1}^{n} x_i \cos((i-1)t)$ for $t \in [0, \pi]$. Show that

$$f(x) = \sup_{t \in [0,\pi]} q_x(t) \; - \; \inf_{t \in [0,\pi]} q_x(t)$$

is convex on $\mathbb{R}^n$. *(Hint: $q_x(t)$ is affine in x for each fixed t.)*

(b) Let $g : \mathbb{R}^n \to \mathbb{R}$ be convex, let $B \in \mathbb{R}^{n \times p}$, $c \in \mathbb{R}^p$, and let $y \in \operatorname{dom} g$. Show that

$$f(x) = c^T x + \inf_{\alpha > 0} \; \frac{g(y + \alpha B x) - g(y)}{\alpha}$$

is convex on $\mathbb{R}^p$.

5. (20 marks)

Let $F_1, \ldots, F_m : \mathbb{R} \to [0, 1]$ be given cumulative distribution functions (CDFs) and let $y \in \mathbb{R}$ be a fixed observed value. For coefficients $a \in \mathbb{R}^m$ set

$$F(u) = a_1 F_1(u) + \cdots + a_m F_m(u), \qquad \operatorname{dom} C = \big\{ a \succeq 0, \; \mathbf{1}^T a = 1 \big\},$$

and let $\omega(u) = 1/(1 + (u - y)^2)$. Define the following function of a:

$$C_\omega(a) = \int_{-\infty}^{y} \omega(u) F(u)^2 \, du + \int_{y}^{\infty} \omega(u) \big(1 - F(u)\big)^2 \, du.$$

Show that C_ω is a quadratic function of a (a quadratic form plus an affine term plus a constant) and C_ω is convex on $\operatorname{dom} C$.

6. (20 marks)

Let $\lambda_1, \ldots, \lambda_n > 0$ and $\kappa_1, \ldots, \kappa_n > 0$ and define

$$f(x) = \prod_{i=1}^{n} \big(1 - e^{-\kappa_i x_i}\big)^{\lambda_i}, \qquad \operatorname{dom} f = \Big\{ x \in \mathbb{R}^n_+ : \sum_{i=1}^{n} \lambda_i e^{-\kappa_i x_i} \leq 1 \Big\}.$$

Show that f is concave on its domain.
(Hint: first set $s_i = \kappa_i x_i$ and prove concavity of the resulting function in s; the substitution acts coordinate-wise, so the concavity inequality transfers back to f without loss. You may use the Cauchy–Schwarz inequality. Consider the boundary of the domain.)